WORKED SOLUTION

STEP II 1996, Question 2

Pure • Nonlinear Simultaneous Equations

STEP II 1996, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q2

The key idea

The equations are not linear in x, y and z, but every term is one of only three products: yz, zx and xy. Treat those three products as the unknowns and the system becomes three *linear* equations, which we can solve by elimination. Multiplying the three products together gives $(xyz)^2$, and dividing xyz by each product in turn recovers the individual variables.

Solution

A linear system in disguise

Put

$$a = yz, \qquad b = zx, \qquad c = xy.$$

Then the three equations read

$$2a + b - 5c = 2, \tag{1}$$
$$a - b + 2c = 1, \tag{2}$$
$$a - 2b + 6c = 3. \tag{3}$$

From (2), $a = 1 + b - 2c$. Substituting into (1):

$$2(1 + b - 2c) + b - 5c = 2 \quad \Longrightarrow \quad 3b - 9c = 0 \quad \Longrightarrow \quad b = 3c.$$

Substituting into (3):

$$(1 + b - 2c) - 2b + 6c = 3 \quad \Longrightarrow \quad -b + 4c = 2.$$

With $b = 3c$ this gives $c = 2$, so $b = 6$ and $a = 1 + 6 - 4 = 3$. The coefficient matrix has determinant $-3 \neq 0$, so this is the *only* solution of the linear system:

$$yz = 3, \qquad zx = 6, \qquad xy = 2.$$

Showing $xyz = \pm 6$

Multiplying the three products together,

$$(yz)(zx)(xy) = x^2y^2z^2 = (xyz)^2 = 3 \times 6 \times 2 = 36,$$

so $xyz = 6$ or $xyz = -6$, as required.

> **Ask yourself** We now know xyz and each of yz, zx, xy. How do we get x on its own?

None of the products is zero, so none of x, y, z is zero and we may divide:

$$x = \frac{xyz}{yz}, \qquad y = \frac{xyz}{zx}, \qquad z = \frac{xyz}{xy}.$$

The possible values

If $xyz = 6$, then $x = 6/3 = 2$, $y = 6/6 = 1$ and $z = 6/2 = 3$.

If $xyz = -6$, then $x = -2$, $y = -1$ and $z = -3$.

Each step above was forced, so these are the only candidates; both satisfy the original equations (the equations involve only products of pairs, so changing the sign of all three variables leaves them unchanged). Hence

$$(x, y, z) = (2, 1, 3) \quad \text{or} \quad (x, y, z) = (-2, -1, -3).$$

Checks

Substitute $(2, 1, 3)$: $yz = 3$, $zx = 6$, $xy = 2$, so the left-hand sides are $6 + 6 - 10 = 2$, $3 - 6 + 4 = 1$ and $3 - 12 + 12 = 3$. All three match. A computer search over all triples of fractions p/q with $|p| \leq 12$, $1 \leq q \leq 4$ found exactly the two solutions above, and exact arithmetic confirms that the linear system has the unique solution $(a, b, c) = (3, 6, 2)$.

Common slips

- Taking square roots of $(xyz)^2 = 36$ separately for each variable, for example writing $x = \pm 2$, $y = \pm 1$, $z = \pm 3$ independently. That gives eight triples, but only two work: the signs are linked because xyz has a single sign.
- Forgetting the negative root, and giving only $(2, 1, 3)$.

Beyond STEP

The trick of renaming awkward expressions so that a nonlinear problem becomes a linear one is used everywhere, from fitting curves like $y = Ae^{kx}$ by taking logarithms to solving differential equations by substitution. Once the system is linear, matrices

and determinants tell you at once whether the solution is unique, which is the starting point of a first course in linear algebra. Strang's *Introduction to Linear Algebra* develops elimination and determinants in exactly this spirit.

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